

DETAILED PROJECT REPORT

AI-Based Government Budget Utilization Monitoring System (GBMS)

INTERNSHIP PROJECT REPORT

Submitted By

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Internship Project Documentation

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1. ABSTRACT

The AI-Based Government Budget Utilization Monitoring System (GBMS) is a full-stack web application designed to monitor budget allocation and expenditure data in a structured and role-aware manner. The platform provides dashboards, department-wise utilization, category analysis, monthly spending trends, expenditure records, alerts, CSV/PDF reports, configurable anomaly thresholds, and administrative audit logs. The application uses Angular, Node.js/Express, MongoDB Atlas, JWT authentication and role-based access control. It is an internship project prototype and does not represent live government transaction records.

2. INTRODUCTION

Public-sector budgeting involves allocation of funds to departments and monitoring of expenditure against approved budgets. GBMS provides a demonstrable centralized software workflow for storing budget and expenditure records, presenting analytics, restricting access by role, detecting rule-based anomalies and generating reports.

3. PROBLEM STATEMENT

Manual or disconnected budget-monitoring workflows can make it difficult to compare allocation with actual expenditure, identify unusual spending patterns, restrict access by responsibility and prepare consolidated reports. GBMS addresses these software-level requirements through centralized storage, role-aware APIs and automated checks.

4. OBJECTIVES

The main objectives are: centralized department, budget and expenditure records; annual and quarterly budgets; Admin, Finance Officer and Department Head roles; utilization calculation; configurable anomaly detection; dashboard analytics; CSV/PDF reporting; audit logging; and browser-based deployment.

5. SCOPE

The system covers authentication, departments, budgets, expenditures, document support, anomaly detection, alerts, dashboard analytics, reports, threshold configuration, user status management and audit logging. It does not directly integrate with government treasury systems or live government financial databases.

6. EXISTING VS PROPOSED SYSTEM

The proposed system replaces disconnected spreadsheet-style monitoring with a centralized MongoDB database, JWT-based role access, automated utilization calculations, rule-based anomaly detection, dashboard analytics, downloadable reports and audit logs.

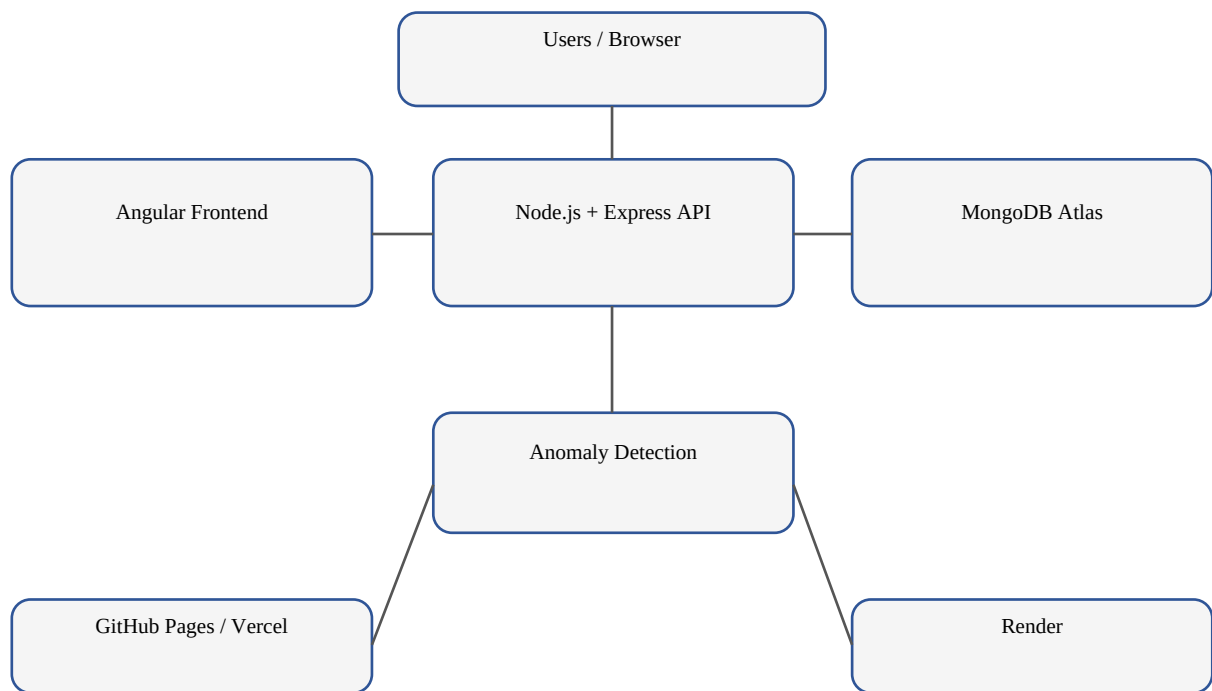
7. FUNCTIONAL REQUIREMENTS

Authentication; department management; annual/quarterly budgets; expenditure CRUD; alert listing and resolution; dashboard summaries and charts; CSV/PDF reports; administrative user, threshold and audit-log management.

8. TECHNOLOGY STACK

Frontend: Angular, TypeScript, HTML and CSS. Charts: Chart.js with ng2-charts. Backend: Node.js and Express.js. Database: MongoDB Atlas with Mongoose. Security: JWT and bcrypt. Version control: Git/GitHub. Deployment: GitHub Pages/Vercel and Render.

9. SYSTEM ARCHITECTURE



The browser accesses the Angular frontend. Angular services call the Express REST API. The backend authenticates requests, applies role-based authorization and accesses MongoDB Atlas. The anomaly detection service evaluates budget and expenditure information and creates alerts.

10. DATABASE DESIGN

User: name, email, password hash, role, department and active status. Department: name, code, type and head. Budget: department, financial year, quarter, category and allocation. Expenditure: budget, department, amount, category and transaction date. Alert: department, budget, alert type, severity, message and resolved state. Threshold: configurable anomaly settings. AuditLog: user, action, entity, details and timestamp.

11. MODULES

Authentication; Department Management; Budget Management; Expenditure Management; Dashboard Analytics; Reports; Alerts and Anomaly Detection; and Admin Management.

12. ANOMALY DETECTION AND ALERTS

The detection service uses configurable thresholds. Demonstration defaults include 40% under-utilization, 70% elapsed-time threshold, 90% overspending warning, 100% critical overspending and a 3× spending-spike multiplier. Rules generate unresolved alerts when their conditions are met.

13. SECURITY AND RBAC

Admin has administrative and financial access. Finance Officer manages financial operations and reports. Department Head is restricted to the assigned department. Backend authorization is enforced independently of frontend guards. Passwords are hashed and JWT protects API requests.

14. REPORTS AND DASHBOARD

The dashboard includes allocated-vs-spent comparison, expenditure by category, monthly spending trend, department utilization, recent expenditures and open alerts. Authorized users can generate CSV and PDF reports, with Department Heads receiving department-scoped reports.

15. DEPLOYMENT

The production frontend points to the deployed Render API rather than localhost. Current project links are: GitHub Repository <https://github.com/PDubey-tech26/GBMS-Mean-Stack> ; GitHub Pages <https://pdubey-tech26.github.io/GBMS-Mean-Stack/> ; Vercel <https://gbms-mean-stack.vercel.app/>

16. TESTING

Login, role restrictions, dashboard, monthly trend, budget CRUD, expenditure CRUD, alerts, reports, admin functions and online deployment were tested during project development and verified successfully in the project environment.

17. LIMITATIONS

The project is an academic prototype. Demonstration/reference data are not actual government transaction records. Anomaly detection is rule-based rather than a trained machine-learning model. Production use would require authoritative int The project is an internship project prototype. Demonstration/reference data are not actual government transac

18. FUTURE SCOPE

Possible extensions include officially permitted government-data integrations, forecasting, statistical/ML anomaly detection, notifications, richer audit analytics, accessibility improvements, automated backups and multi-tenant support.

19. CONCLUSION

GBMS demonstrates a complete full-stack workflow for budget utilization monitoring, combining authentication, RBAC, annual/quarterly budgets, expenditure management, anomaly detection, alerts, analytics, reports and audit logging in one browser-based application.

20. REFERENCES / PROJECT LINKS

GitHub: <https://github.com/PDubey-tech26/GBMS-Mean-Stack> ; GitHub Pages: <https://pdubey-tech26.github.io/GBMS-Mean-Stack/> ; Vercel: <https://gbms-mean-stack.vercel.app/> ; Angular: <https://angular.dev/> ; Express: <https://expressjs.com/> ; MongoDB Atlas: <https://www.mongodb.com/atlas> ; Chart.js: <https://www.chartjs.org/> ; GitHub Pages documentation: <https://docs.github.com/en/pages>